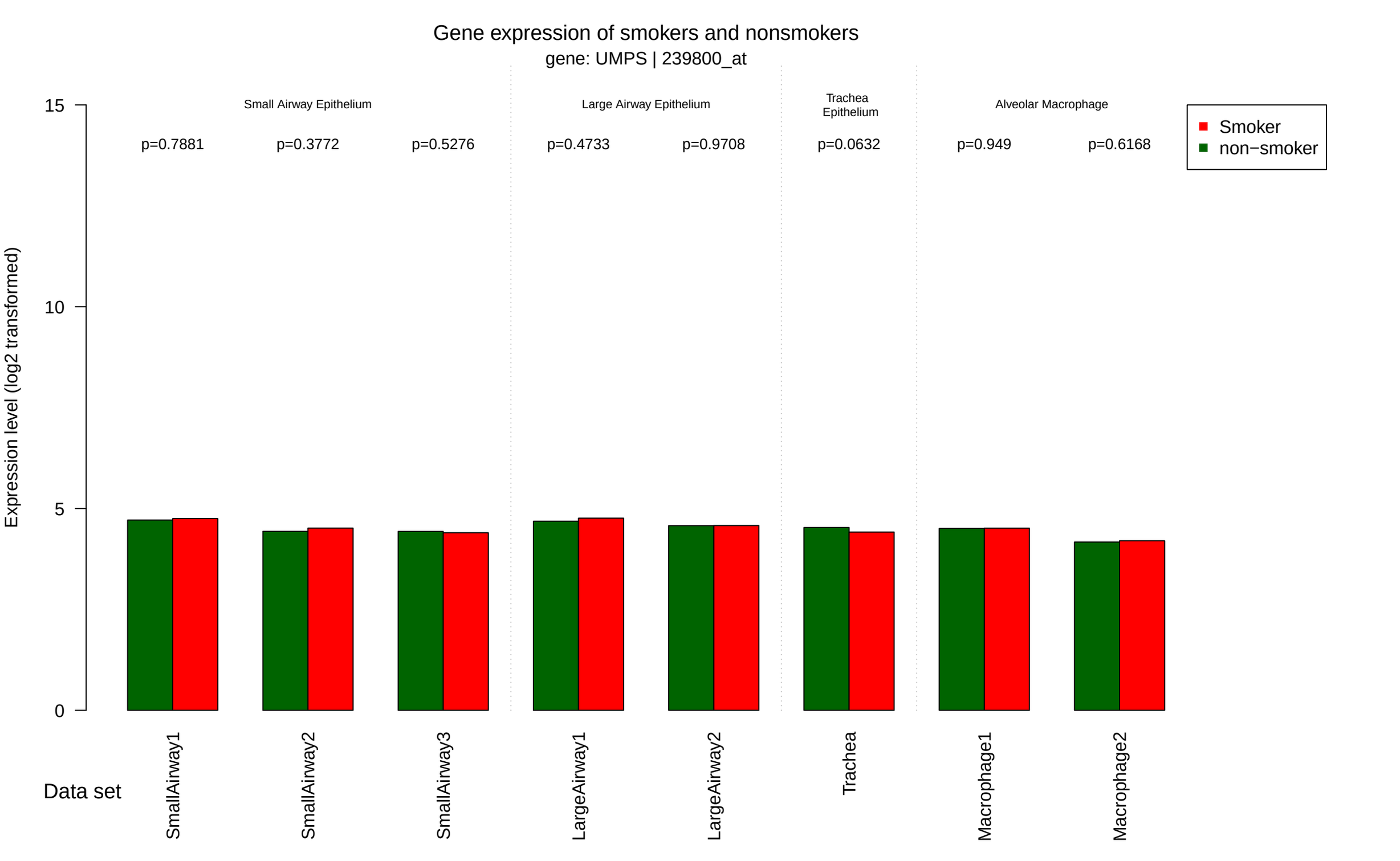
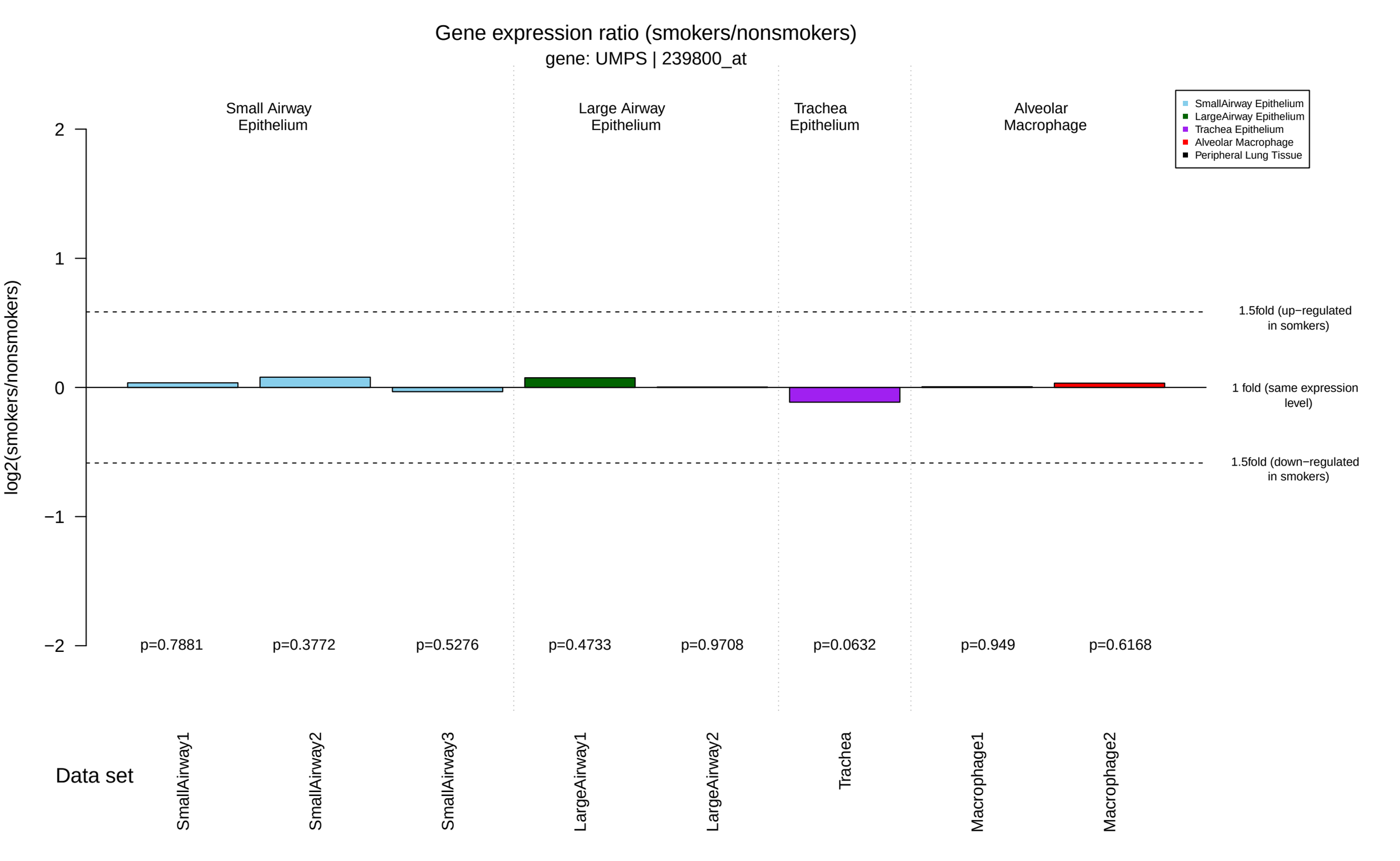
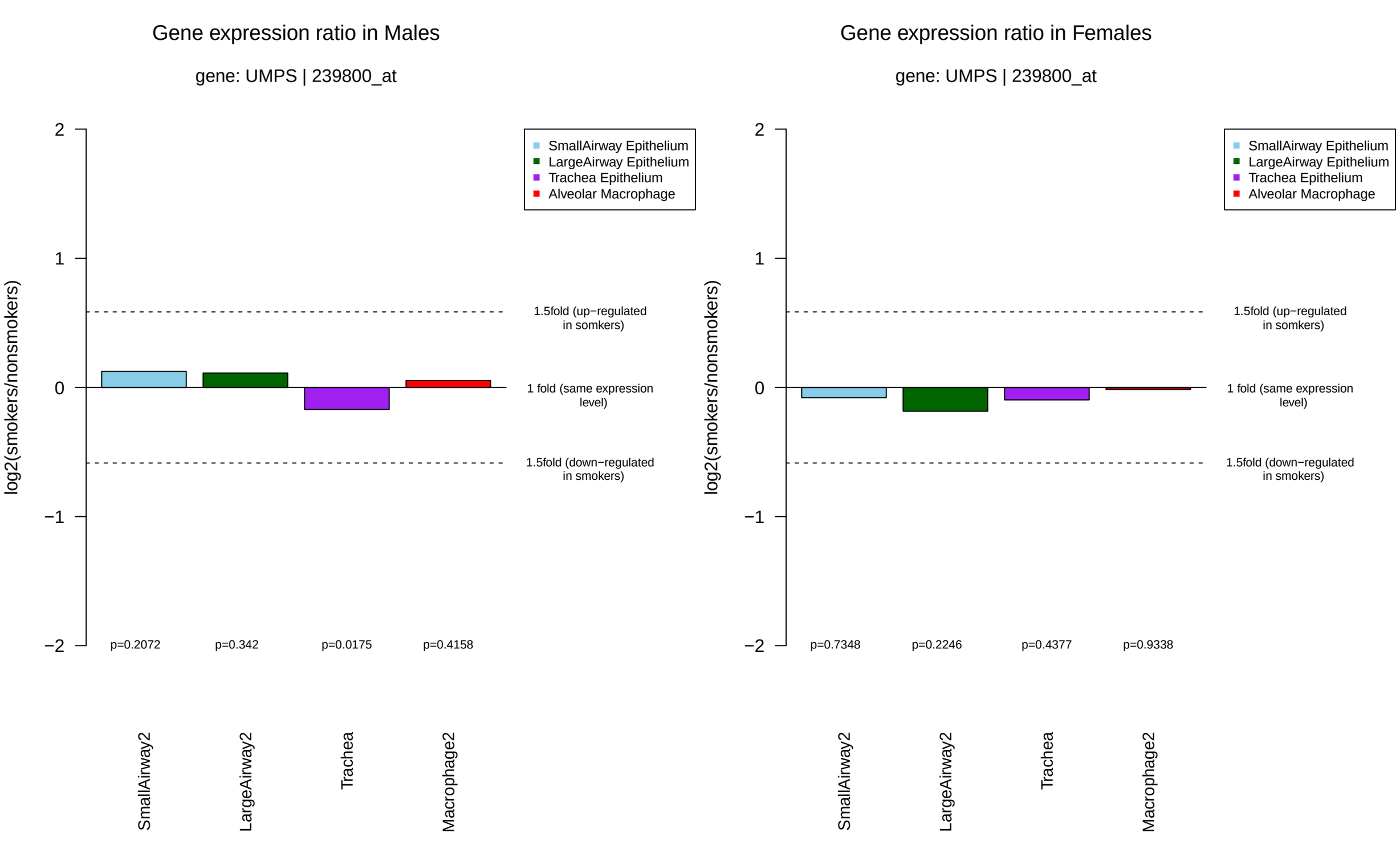
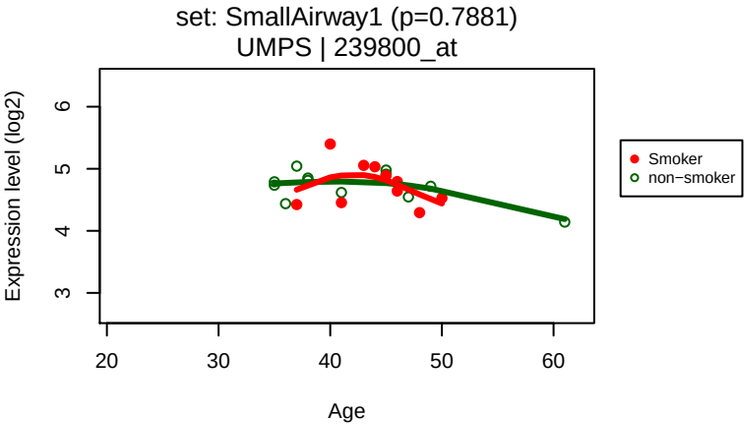
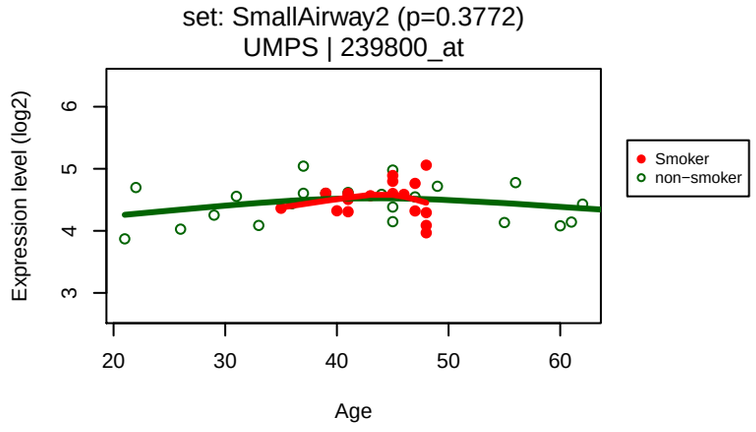
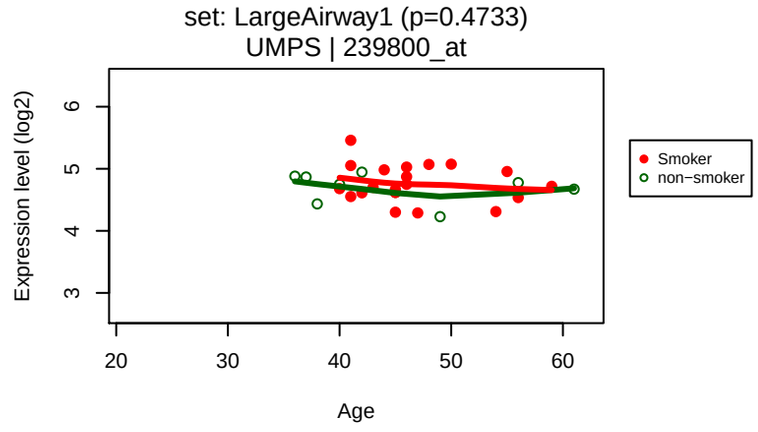

Gene expression of smokers and nonsmokers
gene: UMPS | 239800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7881
p=0.3772
p=0.5276
p=0.4733
p=0.9708
p=0.0632
p=0.949
p=0.6168
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

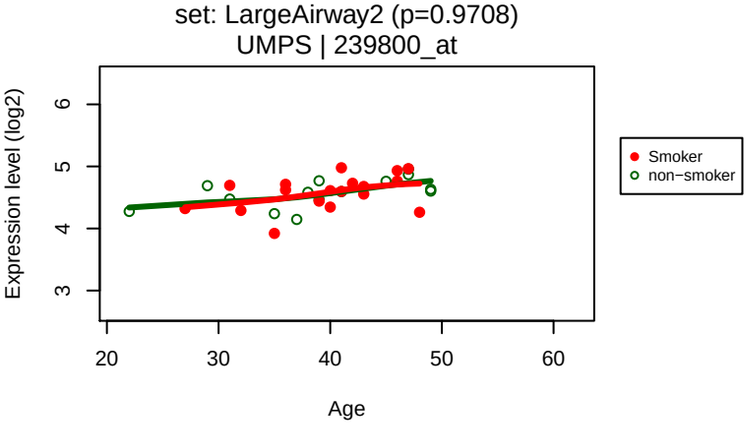

Gene expression ratio (smokers/nonsmokers)
gene: UMPS | 239800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7881
p=0.3772
p=0.5276
p=0.4733
p=0.9708
p=0.0632
p=0.949
p=0.6168
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

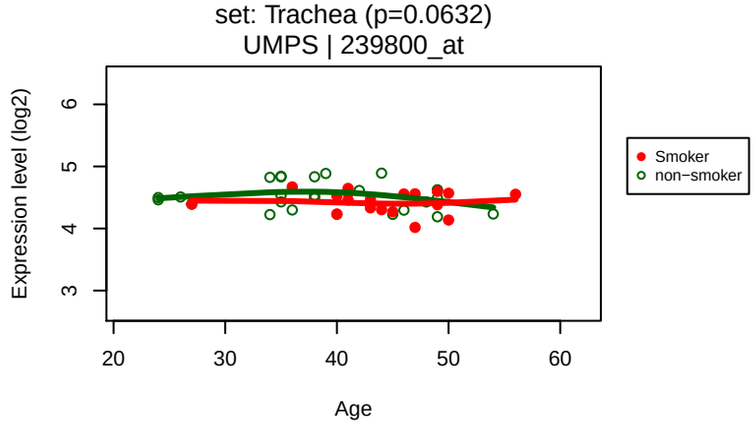

Gene expression ratio in Males
gene: UMPS | 239800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2072
p=0.342
p=0.0175
p=0.4158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: UMPS | 239800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7348
p=0.2246
p=0.4377
p=0.9338
SmallAirway2
LargeAirway2
Trachea
Macrophage2

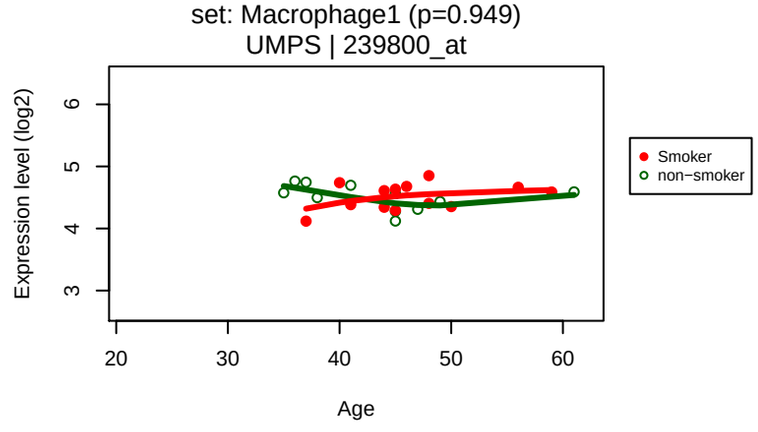

set: SmallAirway1 (p=0.7881)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker

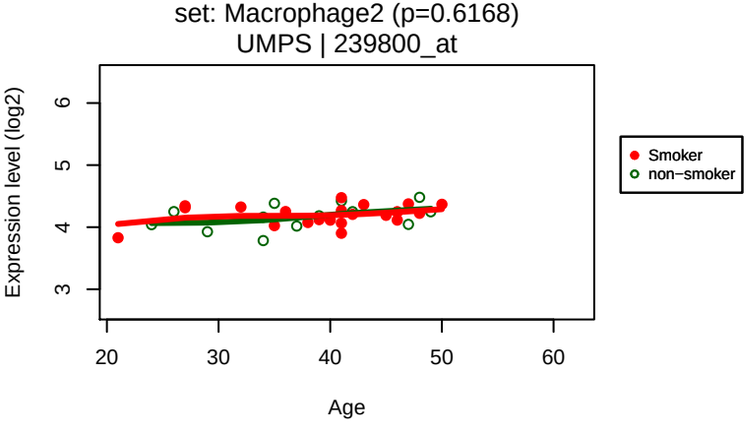

set: SmallAirway2 (p=0.3772)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker

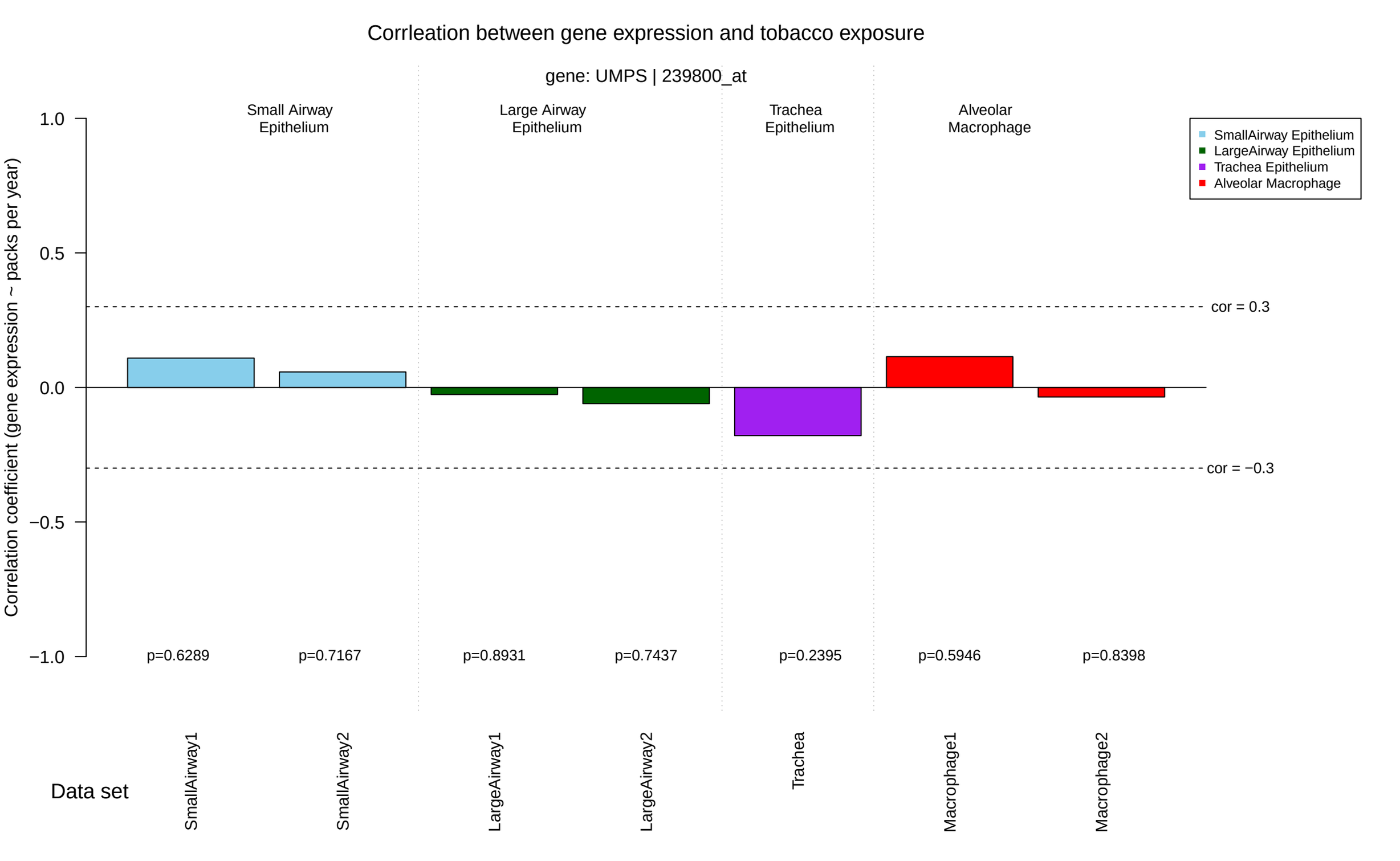

set: LargeAirway1 (p=0.4733)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9708)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0632)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.949)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6168)
UMPS | 239800_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: UMPS | 239800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6289
p=0.7167
p=0.8931
p=0.7437
p=0.2395
p=0.5946
p=0.8398
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2